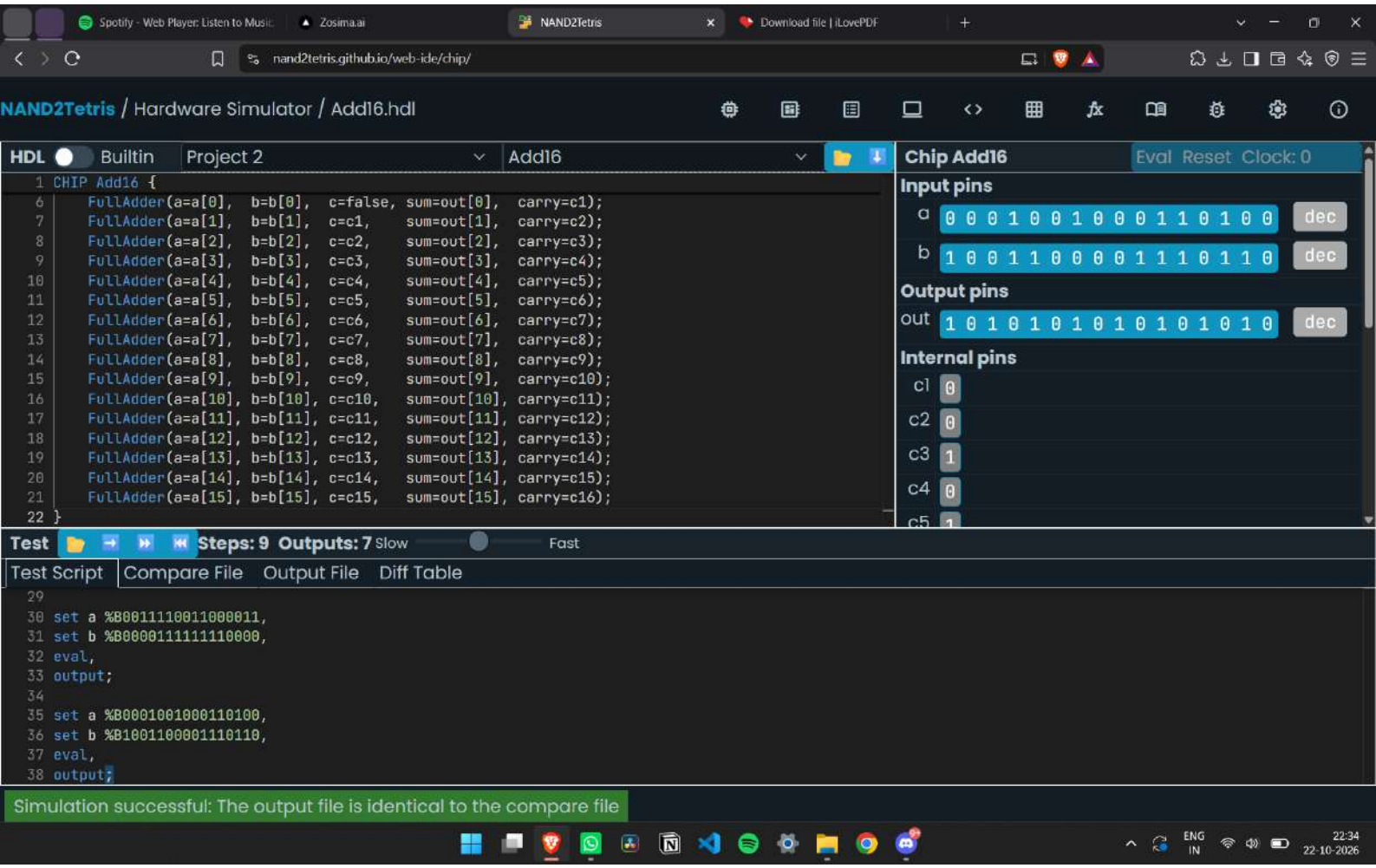

Spotify - Web Player: Listen to Music
Zosima.ai
NAND2Tetris
Download file | iLovePDF
nand2tetris.github.io/web-ide/chip/
NAND2Tetris / Hardware Simulator / Add16.hdl
HDL
Builtin
Project 2
Add16
CHIP Add16 {
FullAdder(a=a[0], b=b[0], c=false, sum=out[0], carry=c1);
FullAdder(a=a[1], b=b[1], c=c1, sum=out[1], carry=c2);
FullAdder(a=a[2], b=b[2], c=c2, sum=out[2], carry=c3);
FullAdder(a=a[3], b=b[3], c=c3, sum=out[3], carry=c4);
FullAdder(a=a[4], b=b[4], c=c4, sum=out[4], carry=c5);
FullAdder(a=a[5], b=b[5], c=c5, sum=out[5], carry=c6);
FullAdder(a=a[6], b=b[6], c=c6, sum=out[6], carry=c7);
FullAdder(a=a[7], b=b[7], c=c7, sum=out[7], carry=c8);
FullAdder(a=a[8], b=b[8], c=c8, sum=out[8], carry=c9);
FullAdder(a=a[9], b=b[9], c=c9, sum=out[9], carry=c10);
FullAdder(a=a[10], b=b[10], c=c10, sum=out[10], carry=c11);
FullAdder(a=a[11], b=b[11], c=c11, sum=out[11], carry=c12);
FullAdder(a=a[12], b=b[12], c=c12, sum=out[12], carry=c13);
FullAdder(a=a[13], b=b[13], c=c13, sum=out[13], carry=c14);
FullAdder(a=a[14], b=b[14], c=c14, sum=out[14], carry=c15);
FullAdder(a=a[15], b=b[15], c=c15, sum=out[15], carry=c16);
}
Chip Add16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0
dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0
dec
Output pins
out 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0
dec
Internal pins
c1 0
c2 0
c3 1
c4 0
Test
Steps: 9 Outputs: 7 Slow
Fast
Test Script
Compare File
Output File
Diff Table
set a %B0011110011000011,
set b %B0000111111110000,
eval,
output;
set a %B0001001000110100,
set b %B1001100001110110,
eval,
output;
Simulation successful: The output file is identical to the compare file
ENG
IN
22:34
22-10-2026